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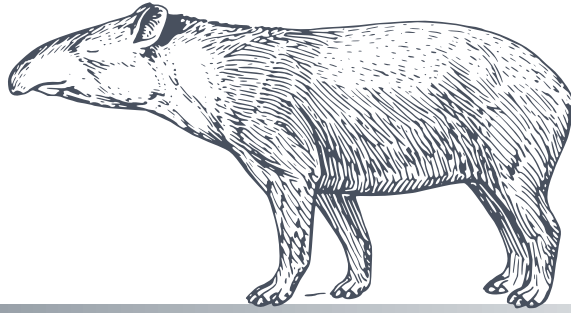
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$$\vec{C} = \vec{A} \cdot \vec{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

1

MATHEMATICAL BACKGROUND

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1.1 PREFACE

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Testing some equations: let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then from the fact that $\mathbb{R}^n$ is a vector space, we get

$$\mathbf{c} = \alpha \mathbf{x} + \beta \mathbf{y} \in \mathbb{R}^n. \quad (1.1)$$

However, if $\mathbf{a} \in \mathbb{R}^m$, where $m \neq n$, then $\alpha \mathbf{x} \in \mathbb{R}^m$, and therefore $\mathbf{c}$ is not defined (i.e. ?? is irrelevant).

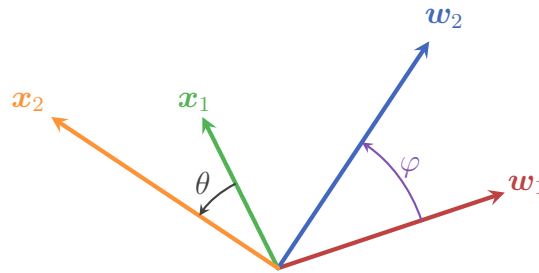


Figure 1.1: This is the first test figure. Does it look ok?

Note 1.1 Test note

This is a test note. I will try to also reference it later.



Now, I wish to reference ??. Did it work?

Also, let us look at an example.

Example 1.1 Some test example

If $x = [1, 2, -3]$ and $y = [5, -1]$, then $n = 3$ and $m = 2$. Clearly, $n \neq m$ - and so $\alpha x \in \mathbb{R}^3$, while $\beta y \in \mathbb{R}^2$. This means that $\alpha x + \beta y$ is not defined.



I believe it is now time for some figures! See ??.

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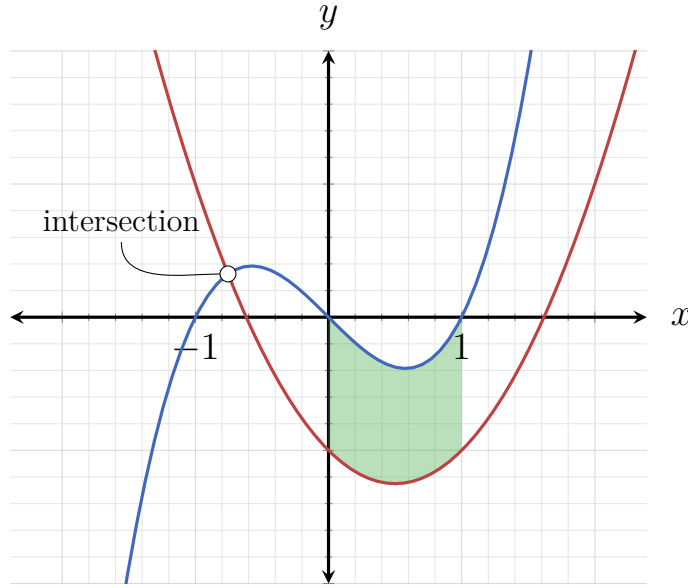


Figure 1.2: And this is another test figure! I hope it works.

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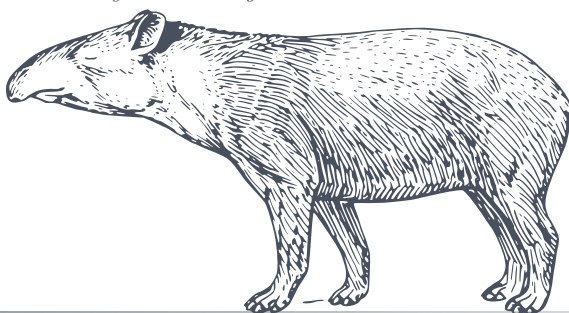
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$$A \vec{x} = \lambda \vec{x}$$

$$(A \cdot B)^T = B^T \cdot A^T$$

$$\det(A \cdot B) = \det(A) \cdot \det(B)$$

$$C = A \cdot B \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

2

THE BASICS OF LINEAR ALGEBRA

Text text text?

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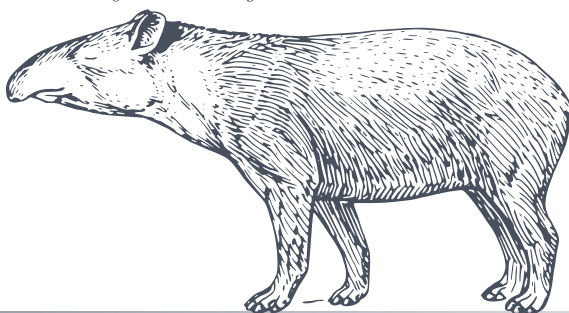
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CHAPTER

3

SOME FORMALISM

Text text text?

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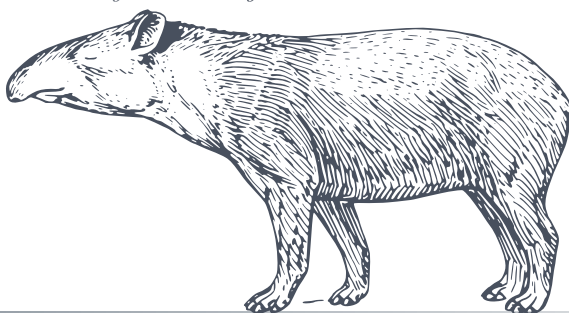
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CHAPTER

4

ADVANCED TOPICS

Text text text?

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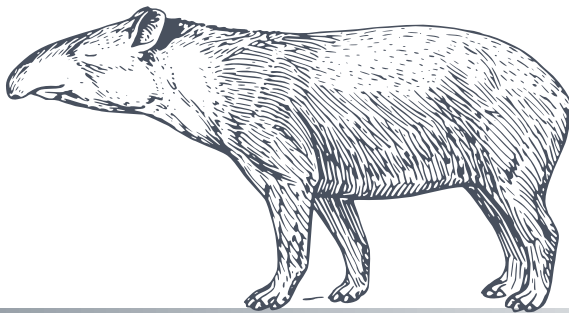
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CHAPTER

5

FURTHER RELATED TOPICS

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CHAPTER 5. FURTHER RELATED TOPICS

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